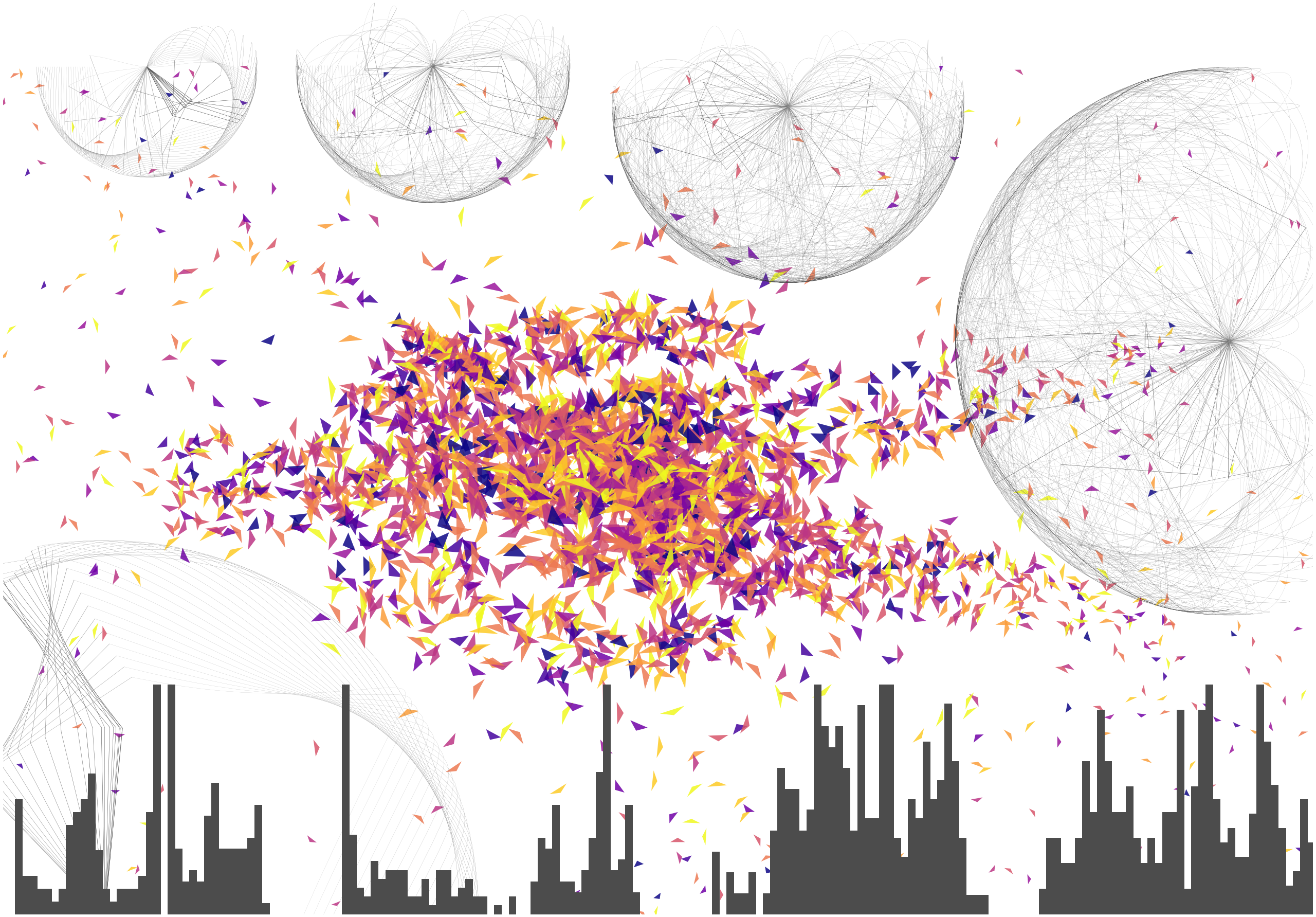
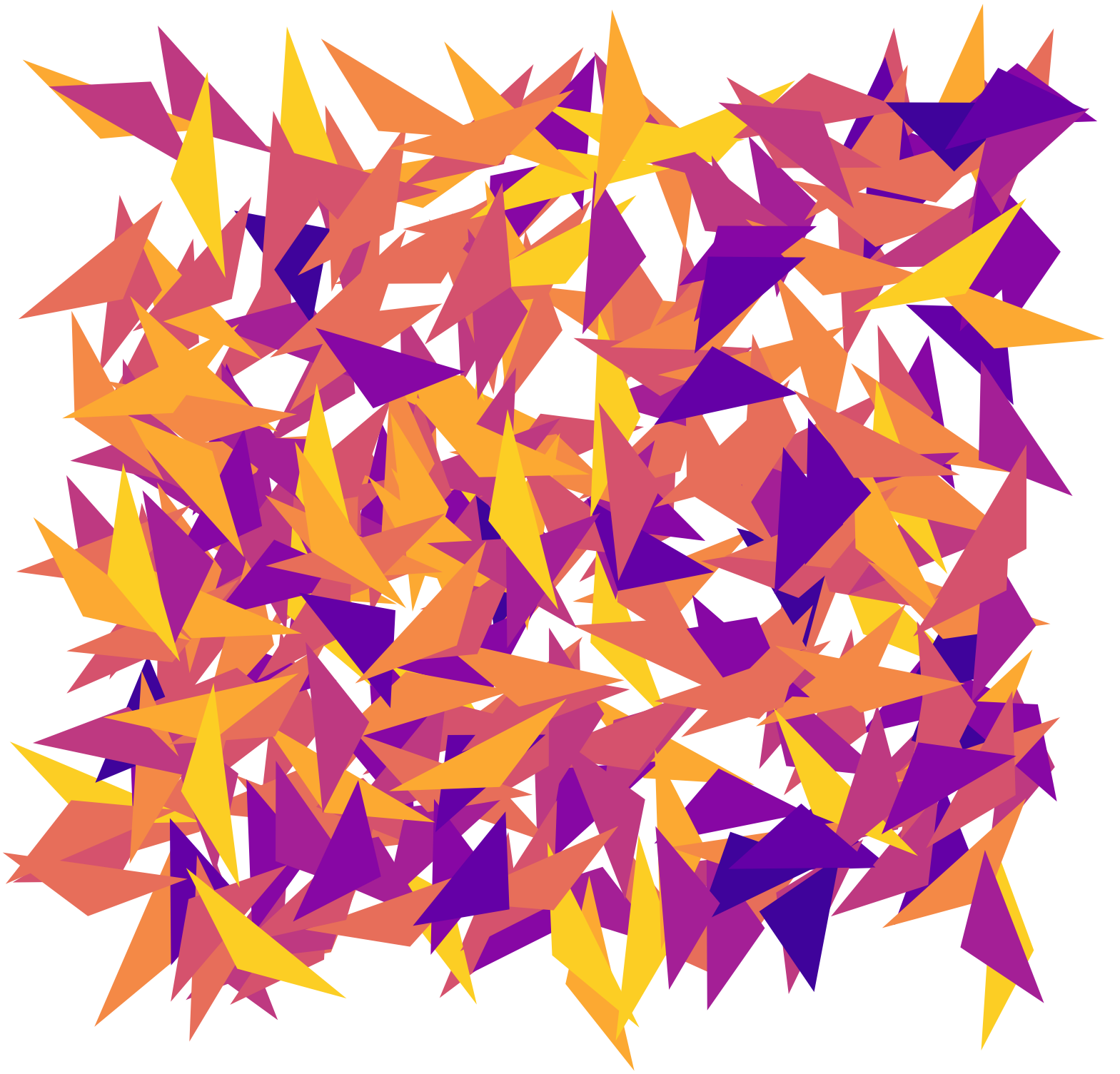
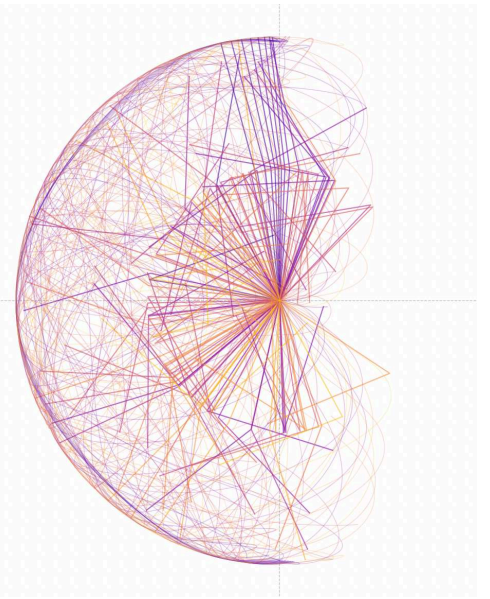
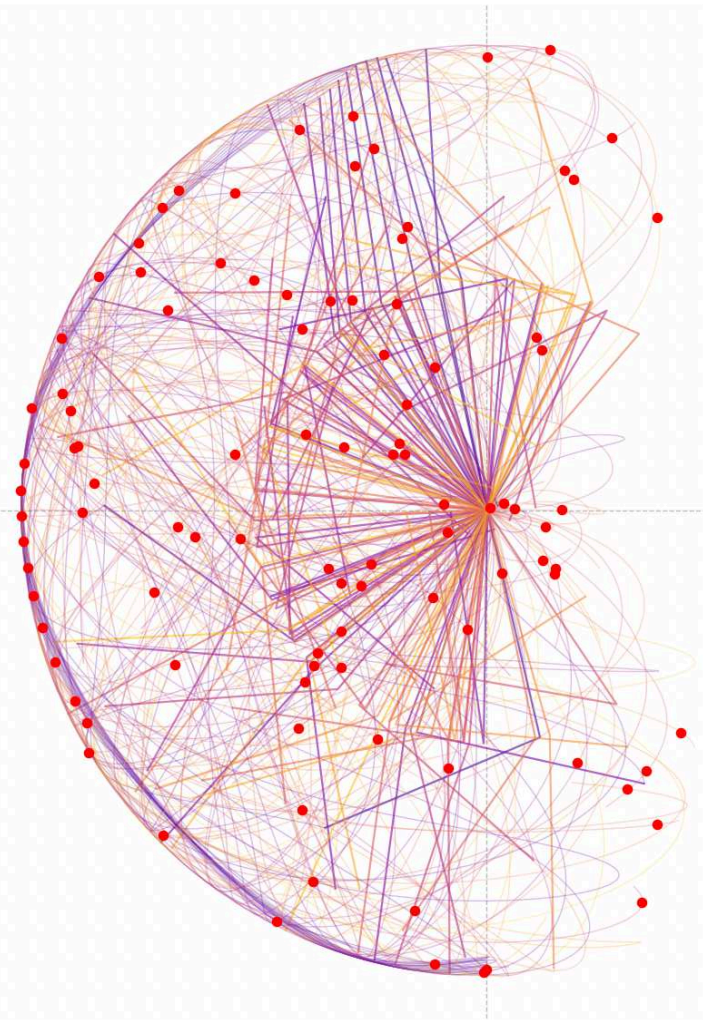
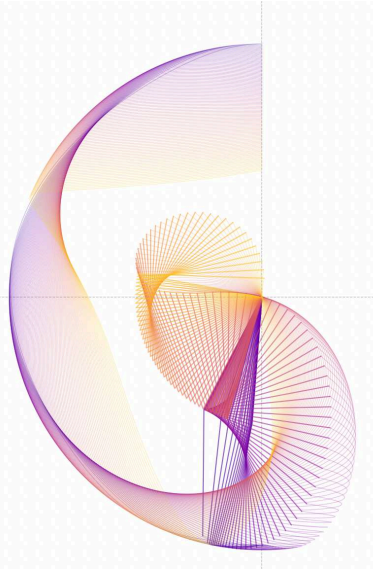
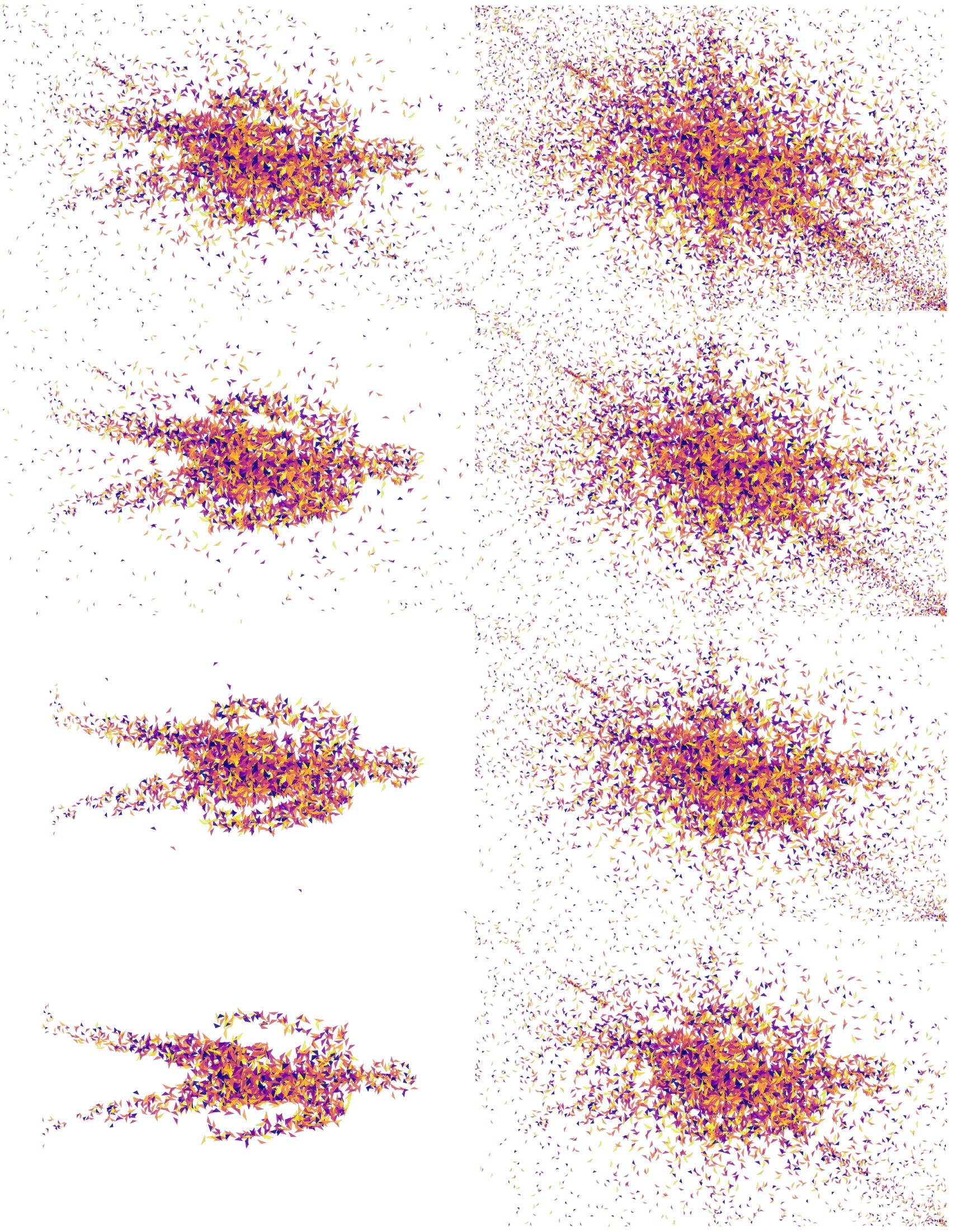


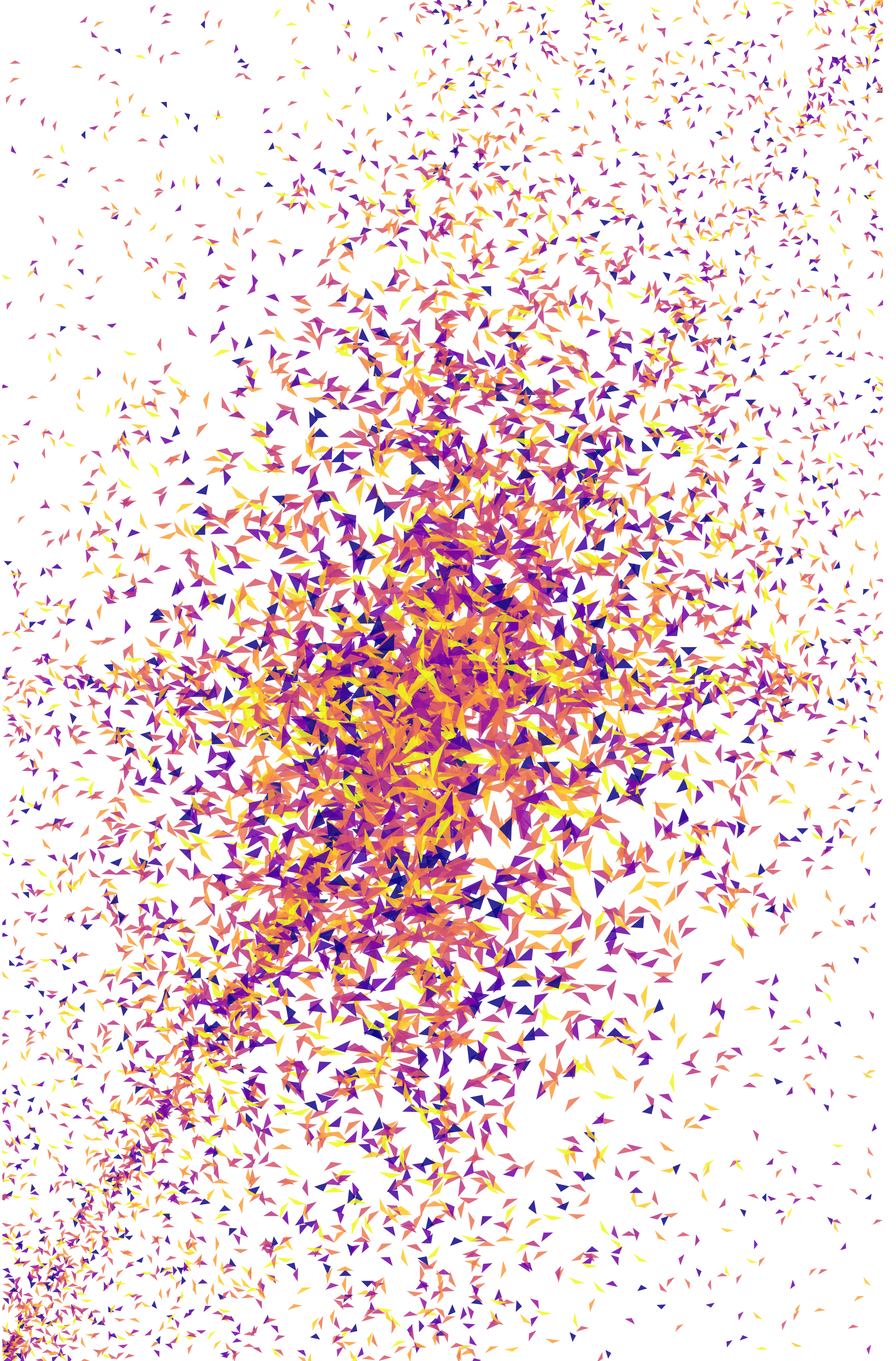


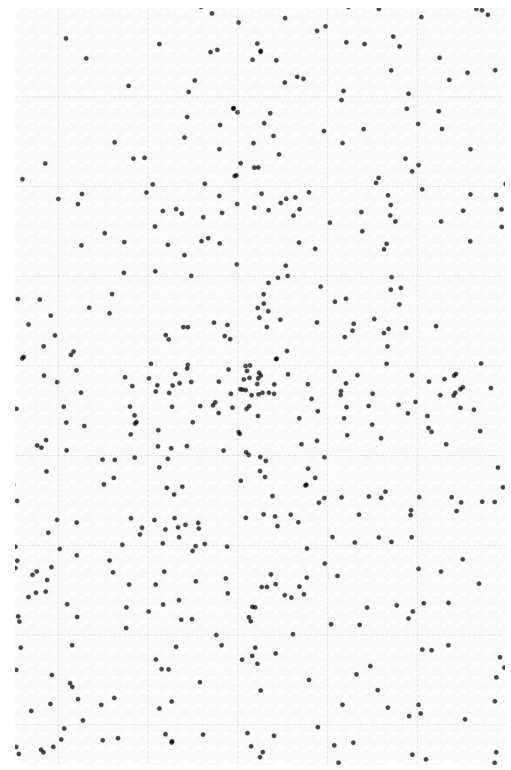


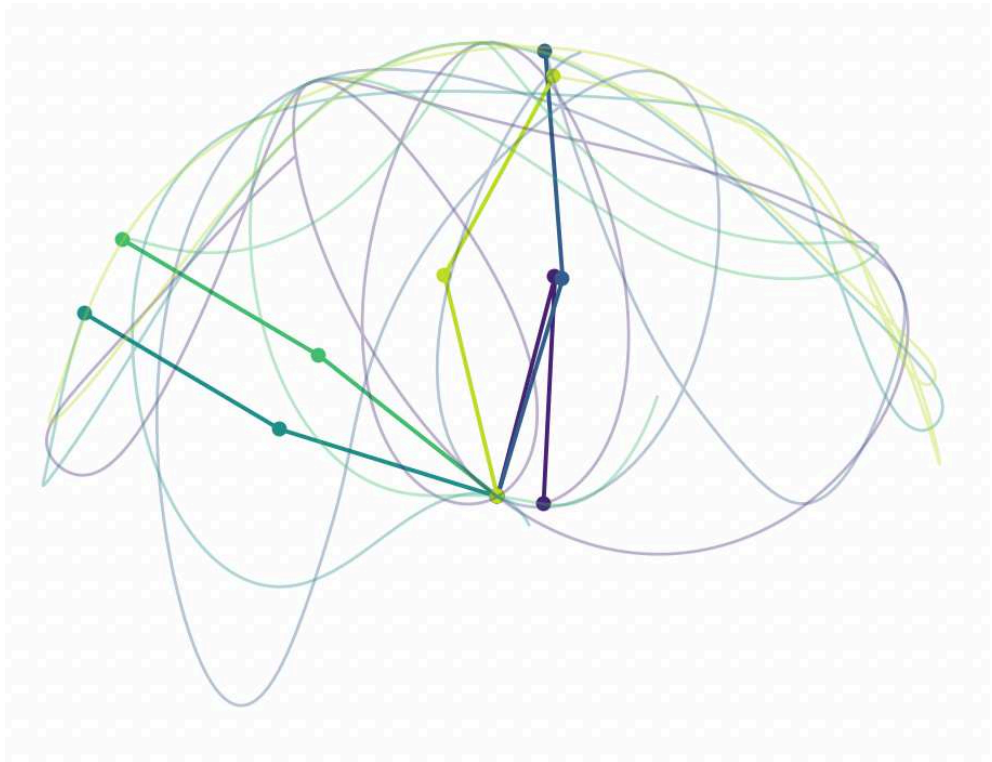


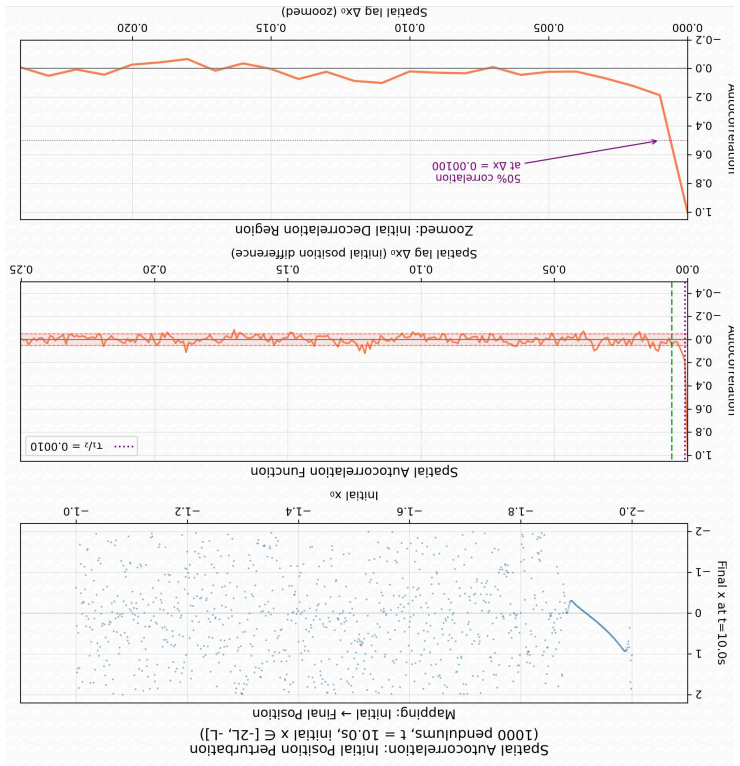


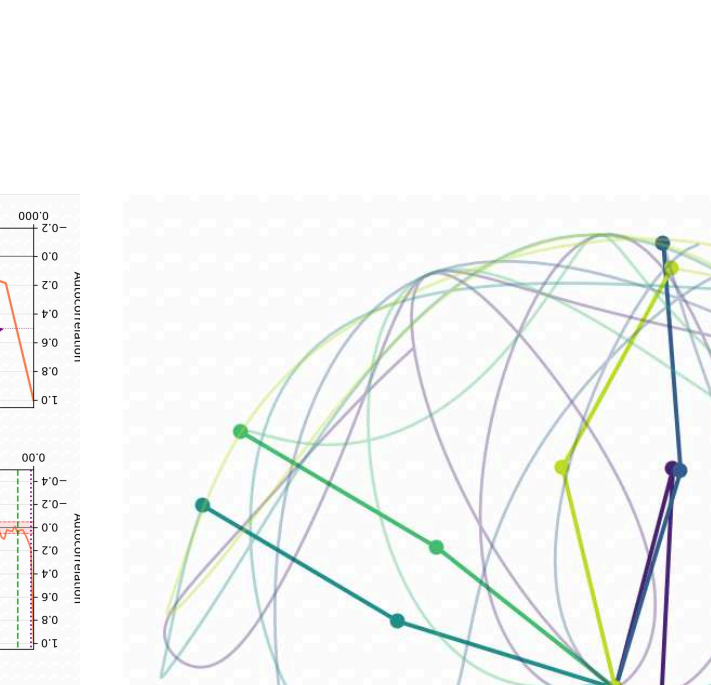









Spatial Autocorrelation: Initial Position Perturbation
(1000 pendulums, t = 10.0s, initial x ∈ [-2L, -L])
Mapping: Initial → Final Position
Final x at t=10.0s
Initial x₀
Spatial Autocorrelation Function
τ₁/₂ = 0.0010
Autocorrelation
Spatial lag Δx₀ (initial position difference)
Zoomed: Initial Decorrelation Region
50% correlation
at Δx = 0.00100
Spatial lag Δx₀ (zoomed)


Double Pendulum: 1000 Initial Positions → t = 10.0s
Initial x uniformly spaced in [-1.8L, -L] at y = 0
Mapping: Initial Position → Final Position
Final x at t=10.0s
Initial x₀
Initial x Distribution (Uniform Grid)
σ=0.231
Count
x position
Final x Distribution at t = 10.0s (Chaotic Spread)
μ=0.139
σ=1.035

Triangle type (0=tall, 9=wide)
y (inches)

Double Pendulum End-Effector x-Distribution Over Time
(1000 pendulums, initial x ∈ [-2L, -L] at x=0)
t = 0s
t = 2s
t = 4s
t = 6s
t = 8s
t = 10s
Count
x position